AI agents as auditors of paleobiology's shared infrastructure: twelve verified cases

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Non-technical Summary

Studies of the fossil record rest on shared resources: online databases of fossil finds, data files published with papers, geological time scales, maps of past continents, and software that estimates rates and ages. An error in one of these resources, or a result that means something different from what its users assume, can carry into the studies that rely on it. While building a research program, two AI systems checked these resources against their sources and against known answers. We report twelve findings they made and confirmed. Each comes with the evidence needed to check it and a practical step that researchers, database maintainers or journals can take.

Abstract

Large-scale paleobiology depends on shared databases, data deposits, time scales, plate models and analysis packages, so errors or misreadings there can affect the analyses built on them. In a research program run by two AI agents with a human facilitator, the agents checked this layer by re-running data transformations, software and published calculations against known answers and their own sources. We report twelve cases, each rerun from fixed, checksummed inputs and checked by a second agent. They include deposited records absent from a dated public database snapshot, interval ages that change between database versions, defects in age–depth conversion and data-cleaning software, simulated preservation-rate intervals that miss the true value too often, plate models that disagree on paleolatitude by a median of 2.9–13.6°, and a printed probability of 3.4×10^{-16} that its stated test puts at 0.32. Review corrected the initial accounts of six cases. AI agents can audit this shared layer systematically, and each finding gives researchers a specific, verifiable action.

Introduction

A diversity curve, an extinction rate or a paleolatitudinal gradient is the last link in a long chain. Fossils become occurrence records in databases such as the Paleobiology Database (PBDB) and in data deposits published with papers. A geologic time scale and age models place the records in time, plate reconstructions place them in space, and analysis packages turn them into estimates (Peters and McClennen 2016; Gradstein, F. M., J. G. Ogg, M. D. Schmitz, and G. M. Ogg, eds. 2020). Global diversity studies show how such a chain is assembled, from database occurrences through explicit choices about preservation, sampling and analysis (Alroy et al. 2008). Each link is maintained by others, and each changes over time.

Paleobiologists have long studied how the rock record distorts inference through sampling, preservation and the availability of rock (Smith 2001; Dillon et al. 2023). The shared infrastructure adds a second set of dependencies, of four kinds: the transformations that turn sources into data products, the software that implements each calculation, the models chosen where several are defensible, and the meaning of what each tool reports. These dependencies lie between the rocks and the result in every large-scale analysis.

The community has named parts of this layer as priorities. Among 89 "Big Questions" in paleontology are questions on finding and correcting data-entry bias, recovering lost historical information, reporting, and integrating data (AFR2–AFR4, FI1) (Smith et al. 2025). Guidance on computational reproducibility asks researchers to record how each result was produced, keep versions and inspect intermediate results (Sandve et al. 2013).

Checking the layer means comparing data with their sources, running software against known answers, inspecting code and recomputing published numbers. The work is broad and exacting. In this program, two AI agents did it and recorded the evidence, command and input hashes for each check. A benchmark of agents reproducing the computations of published papers is a relevant precedent (Siegel et al. 2024).

Here we report twelve findings made by two AI agents while they built a paleobiology research program in October and November 2026. Three questions organize them: where each finding arose, what checks established it, and what changed when a second agent re-examined it. The cases show agents auditing the shared layer at the scale of a whole program, with findings specific enough for researchers, maintainers, journals and other agent-based programs to act on.

Methods

Setting

The Extinction and Paleontology Research Programme (EPRP) began in October 2026. Two AI agents, Anthropic's Claude and OpenAI's Codex, run it with a human facilitator (B.W.B.), who sets direction and holds every external decision, including contact with data providers and authors, release and submission. The agents work in a public repository and coordinate through a written correspondence board. Their contributions are listed by system, version and task under Roles of AI Systems.

How Candidate Findings Arose

The program's first task was to vet research questions against the data and software that could answer them. To do so, the agents:

- mirrored databases, data deposits and plate models, recording a SHA-256 checksum for every file on arrival, and queried the PBDB through its public interface (Peters and McClennen 2016);
- benchmarked analysis packages against simulations with known answers, in validation tests declared before they were run (with dated amendments and recorded deviations) and in exploratory follow-up probes;
- recomputed numbers printed in published papers from their stated inputs and methods.

Three working sessions, for databases, packages and references, logged each finding with its evidence and the checks it had received. The resulting catalogue held 189 initial candidate findings, including positive controls and intended behaviors; later findings joined as the program continued.

Selection and Coding

On 8 November 2026 we selected twelve cases for their consequence for inference (what each would change in an analysis someone plausibly runs), their variety, and the inclusion of positive controls. Licensing problems, bibliographic metadata and findings involving sensitive location data were excluded. Supplement S1 gives the selection record and its reserves.

On 12 November we coded each case by the family of check that established it. The four families follow the four kinds of dependency and are defined by what was compared:

- **data provenance and transformation (DP):** membership or values compared across identified sources, versions or derived products;
- **computation and implementation (CI):** retained or published numbers compared with a specified calculation under identified reference inputs;
- **model sensitivity (MS):** named alternative models compared on shared inputs;
- **output interpretation and calibration (OI):** the quantity or uncertainty process a reported output targets, and its operating characteristics in a stated test design.

Each case has one primary family and, where needed, a secondary one. Its discovery route comes from dated records in the repository and correspondence (Table 1).

Replay and Checks

On 8 November 2026 Codex's replay team (the main Codex session and GPT-6.1 Sol subagents) replayed each case from pinned inputs (the exact files used, identified by checksum), by fresh software calls, recalculation from source tables or reaggregation of retained outputs. Cases 11 and 12 were first recorded by Codex's reference and package sessions; their November checks added a complement recurrence and integer-polynomial convolution (case 11) and a separately written scorer (case 12). Where possible, key numbers were rederived by a separate route: independent aggregation, interpolation, geometry or scoring code, recounts from raw files, or closed-form arithmetic. Each verified proposition entered a claims ledger with its value, evidence files, verifying commit, reproduce command and scope. The ledger holds 22 propositions, and every number in the Results is tagged to one of them.

Review and Record

A wording review checked every draft sentence against an interpretation specification that states what each case establishes. Corrections at each stage were logged with their finder and replacement (Supplement S3). Versions, environments and input hashes accompany each replay in the public repository, together with reviewed snapshots of the correspondence. A tagged snapshot will be archived with a DOI at release.

Results

Overview

The agents' checks reached links all along the chain, from deposit to published number (Fig. 1A), and made findings in each of the four families (Table 1): five cases of data provenance and transformation, four of computation and implementation, two of output interpretation and calibration, and one of model sensitivity. The GRIWM case also carries a computation finding. Every case was replayed from pinned inputs and checked a second time (Fig. 1B). We present the cases by resource, as a reader meets them.

The Paleobiology Database

A published global analysis of marine latitudinal diversity (Benson et al. 2025) deposited its occurrence data. Of the 368,760 occurrences in the deposit's molluscan member, 71,103 were absent from a public PBDB snapshot taken on 23 October 2026, and 297,657 were present (Fig. 2A). The difference reached our own analysis: a presence screen for tropical support outside Europe and North America passed in 14 of 21 stages with the full deposit and in 12 with snapshot members only. We report these records as counts.

Archived PBDB query products show interval ages changing between snapshots. Across three bracketed snapshot pairs, 460, 317 and 17 intervals changed their modal base age (the most common base age among their records), counting intervals with at least 50 single-interval occurrences in the later snapshot (Fig. 2B); applying the threshold to both snapshots changes the middle count to 316. The database serves base and top ages, in Ma, as fields of each interval record (Paleobiology Database 2026), so the same query run on different dates can return different ages for the same interval. Recording the date and version of each query makes the change visible.

Microfossil Databases

The 2023 update of the Triton planktonic foraminifera database (Fenton et al. 2021) contains 28,832 excess rows under the key we specified (Supplement S2). A conservative recipe removes lower-priority-source rows from cross-source groups with recorded depth, keeping every row from the highest-priority source; it removes 16,260 rows. One of our own analyses averages each lineage's absolute paleolatitude within each time bin; the removal changed 1,388 of 2,267 comparable means (Fig. 2C), 411 of them by at least one degree. Lineage-pair classifications changed in 8, 8, 2 and 1 of 491 pairs under four classification rules.

The final Neptune Sandbox Berlin (NSB) database (Renaudie et al. 2020) contains 61,862 sample records and, for each drill hole, a set of depth–age tie points. We recomputed each sample's age by linear interpolation between its hole's tie points, taking the first age where two tie points share a depth. Of 56,387 samples with both a stored and a recomputed age, 4,111, in 56 holes, differ by more than 0.01 Myr (Fig. 2D). Taking the last age at shared depths instead removes 8 of these differences. A user who takes the stored ages and one who recomputes them from the tie points work from different numbers.

Analysis Packages

admtools converts stratigraphic height to time through an age–depth model (Hohmann 2023; Hohmann et al. 2025). We tested version 0.7.0 on a synthetic model with two hiatuses, against exact linear-segment truth. With sparse tie points, it mis-converted 3,972 of 8,282 heights, by up to 3.0 Myr (Fig. 3A). With dense tie points, it mis-converted 6 of the same heights, by up to 0.466 Myr, and a targeted probe near a hiatus gave an error of about 2.7 Myr. The errors lie in the interpolation segments adjoining the hiatuses.

fossilbrush flags occurrences whose ages fall outside a reference range by at least a threshold in Myr (Flannery-Sutherland et al. 2022; Flannery-Sutherland et al. 2022). Version 1.0.6 compared those differences with the threshold as text. At the default threshold of 5, it left differences of 10, 20, 49.9, 100 and 499 unflagged while flagging 50 and 99; character order predicted every decision on the test grid (Fig. 3B).

The Gaussian-resampled, inverse-weighted McInerny method (GRIWM) estimates extinction time from dated records (Bradshaw et al. 2012). Its implementation in Rextinct 1.1.0 overwrote the caller's random-number state: after a call, the calling program's random stream was the same whatever seed it had set beforehand. Any analysis that draws random numbers after the call continues from a stream the package chose.

Published Products

Four published products differed from their sources or stated tests.

- **Capture histories.** A study of extinction selectivity (Monarrez et al. 2023) deposited its capture histories (each genus's record of presence and absence through time) both as an intact binary file and in a spreadsheet. 2,967 of the 6,562 spreadsheet histories differed numerically from the binary deposit. A further 3,595 are restored exactly by padding to the known width of 86 characters.
- **Fossil Record 2.** The one-file All-Life table of the digital Fossil Record 2 compilation (Benton, M. J., ed. 1993) held 220 rows whose cell tails were shifted one column left of their chapter workbooks: 112 shift ranges, 2 shift habitat and range, and 106 move a not-found-in-record flag into the Holocene column.
- **A printed probability.** An analysis of cyclicity in stratigraphic boundaries (Rampino and Caldeira 2025) states a binomial test with n = 47, k = 28 and p = 0.55 and prints a tail probability of 3.4×10^{-16}. The stated test gives 0.316, by exact rational arithmetic, integer-polynomial convolution, a direct exact sum and R.

- **Archived calibration scores.** INDITEK-2.0 (Herrero-Gascón et al. 2026) deposits a synthetic demonstration with archived calibration proposals. Rerun through the pinned harness, three rejected proposals reproduced their prediction vectors to within about 5×10^{-14} across 2,978 cells. Their saved residual sums of squares differed from rescoring under the deposited observations by 42.4, 17.5 and 11.9, and a separately written, model-free scorer agreed with the rescored values.

Model Choices and Output Meaning

We simulated fossil records under a declared constant-rate, homogeneous-Poisson design and asked how often the 95% intervals for the preservation rate from PyRate, a Bayesian program for estimating speciation, extinction and preservation rates (Silvestro et al. 2014), covered the true value. They covered it in 81 of 100 replicates in one arm and 33 of 50 in a lower-preservation arm (Fig. 3C). Posterior medians sat above the truth, with median relative bias of 0.0895 and 0.458, and every miss fell below the lower endpoint: the intervals were shifted high.

The same benchmark carried positive controls for a different target: confidence bounds on the true end of each retained species' range, in two simulated clades. Under homogeneous sampling, the classical Strauss–Sadler bound (Strauss and Sadler 1989) covered the truth in 1,895 and 1,899 of 2,000 cases in two implementations, and the Robson–Whitlock bound (ROBSON and WHITLOCK 1964), as implemented in EDE 0.2.0, in 1,909 (Fig. 3C). In a hiatus arm, Strauss–Sadler coverage fell to 1,125 of 1,272 retained cases.

GRIWM's interval in Rextinct 1.1.0 comes from resampling the input dating errors, and with zero dating error it collapses to a point. It therefore measures how sensitive the estimated extinction time is to uncertain dates, given the records (Bradshaw et al. 2012).

We reconstructed 300 public occurrence points at six ages in three plate models: MERDITH2021 (Merdith et al. 2021), PALEOMAP (Scotese 2016) and MATTHEWS2016_pmag_ref (Matthews et al. 2016). Over the points reconstructable in all three models at each age, the median spread in paleolatitude was 2.9° at 10 Ma, 5.7° at 50 Ma, 4.3° at 100 Ma, 5.5° at 200 Ma, 4.7° at 300 Ma and 13.6° at 400 Ma (Fig. 3D); 194 points are reconstructable at every age. Two routes to the TorsvikCocks2017 model (Torsvik and Cocks 2016) differed by an approximately rigid, age-specific rotation, consistent with the two reference frames its authors distinguish (Supplement S2).

Checks and Review

Each finding rests on its own mix of re-execution, recomputation from source, code inspection and arithmetic (Table 1). The INDITEK case paired a forward replay through the original harness with separately written rescoring; the printed probability was confirmed four ways. The recorded replay and review trail corrected the initial accounts of six cases (Table 1; Supplement S3):

- PyRate's intervals moved from "too narrow" to "shifted high";
- admtools errors moved from "above every hiatus" to the adjoining segments;
- the Triton counts of changed means were recomputed under the pinned recipe;
- PBDB records described as "withheld" became records absent from a dated snapshot;
- a count of spreadsheet cells in scientific notation was separated from the count of altered histories;
- the NSB sample count was corrected for a header row.

Discussion

Systematic checks by two AI agents turned up verifiable findings in all four kinds of dependency on which large-scale paleobiology relies. Each case changes what a researcher should record, test or interpret, and each comes with its evidence, pinned inputs and command. Because the findings concern named versions of named resources, they are actionable: a maintainer has a specified check to run, and a user can test whether a finding touches their own analysis.

Comparison with a Source

Where a source or known answer is specified, a direct comparison establishes a discrepancy in the records present, interval ages, row counts, conversions, flags or published numbers. The comparison needs the query date, the version and the intermediate products, so recording them lets a researcher test inputs before fitting any biological model (Sandve et al. 2013). GBIF builds this into its downloads: each records its query predicate and receives a DOI that identifies the extract (GBIF Secretariat 2026*a*, *b*). Locating a discrepancy and explaining its cause are separate steps, and the first is what makes the second possible.

Model Sensitivity and Interval Meaning

The three plate models placed the same points degrees apart, so an analysis whose result depends on paleolatitude should state its model and cohort and report the spread among alternatives. Published paleolatitude tools propagate uncertainty in paleomagnetic poles and ages, and treat the choice of reference frame as a separate decision (van Hinsbergen et al. 2015); our comparison measures sensitivity to the choice of reconstruction model. An interval calls for two further judgments: which uncertainty process it propagates, and how often it covers the truth. GRIWM's interval propagates dating error, and PyRate's preservation-rate intervals covered the truth less often than nominal in simulation. Both kinds of finding change how a result should be read.

Actions

Each finding points to a step that a specific audience can take (Table 2). Most are small changes to existing practice, such as recording a query date, comparing numbers as numbers or archiving a scorer with its scores.

The Agents' Own Record

The audit applied to the auditors too. Replay corrected six initial accounts, and wording review corrected this paper's first draft, which had joined a count from one admtools control with the maximum from another and attributed the INDITEK archive to the wrong paper (Supplement S3). The same source comparisons and known-answer checks could serve other fields built on community databases and shared code. Biodiversity informatics is one: taxonomic checks and coordinate validation against reference distributions have long been part of its data-quality practice (Yesson et al. 2007). We offer that application as a proposal.

Limitations

- **Selection.** The cases were selected for consequence, so they describe kinds of finding, not their frequency in the infrastructure.
- **Downstream scope.** We replayed published calculations and part of one deposited model, not whole downstream studies; measured consequences are to support counts, inputs and our own analyses.
- **Independence.** Checks vary in independence. The INDITEK forward run reused the original harness, and some checks shared inputs or model family.
- **Calibration controls.** These rest on two simulated clades. One declared agreement gate failed: it required at least 0.99 of paired range-end bounds to agree within 0.02 Myr, and 0.91 did. The executed hiatus windows were shifted one Myr from the declared design.
- **Causes and status.** The causes of the PBDB absences, the NSB disagreements and the INDITEK score mismatch are unresolved. Neither the stored nor the recomputed NSB ages are certified, and the plate-model spread is structural disagreement among models, not calibrated error.
- **Reproduction.** Some inputs (container images, posterior logs, source archives) are held locally, and a cold rebuild from public inputs alone is untested.
- **Disclosure.** Providers and authors had not been contacted at the time of writing, and whether they already knew of a finding is unresolved.

Conclusions

AI agents can systematically check the databases and software behind fossil-record research, and in doing so they found verifiable errors and pitfalls that researchers need to know about. Three studies would extend this

case series, each with its own payoff. An audit of a defined population of dependencies would estimate how often such findings occur; a matched comparison of agent and human auditors would measure their relative performance; and replay of downstream studies would show what particular findings change.

Roles of AI Systems

- **Codex (OpenAI)**, with GPT-6.1 Sol subagents, took over the program's database, reference and package work from 28 October 2026; first recorded cases 11 and 12; replayed all twelve cases; built the claims ledger; reviewed this paper's wording and reconstruction; and checked its cited sources.
- **Claude (Anthropic)**, Opus 5.5 in the main session with Sonnet subagents, ran the database, package and reference sessions until 28 October; compiled the 189-row catalogue; selected the twelve cases; wrote the interpretation specification; and drafted this paper.
- **Barry W. Brook** facilitated the program, endorsed developing the paper idea on 25 October 2026, chose the case series for development on 8 November 2026, and holds decisions on contact, submission and release.

In line with *Paleobiology*'s policy, the AI systems are not authors. Their analytical roles are described in Methods and their role in writing the text here; any role in producing a figure will be stated in its caption.

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Data Availability Statement

The repository is public at https://github.com/BWBrook/eprp; code is MIT-licensed, and text, figures and data are CC BY-SA 4.0. The claims ledger (`papers/infrastructure-audit/manuscript/claims.csv`) gives each proposition's evidence files, verifying commit and reproduce command. Third-party sources keep their own licenses. Records absent from public database snapshots are reported as counts only. A tagged release will be archived on Zenodo with a DOI.

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Figure Captions

Figure 1. The checked dependencies and the audit workflow. **A**, The twelve cases placed along the path from rocks to a published result (collection, database or deposit, time scale and age model, plate model, analysis package, published number), colored by primary family of check. **B**, The workflow: 189 initial candidate findings and later additions, twelve selected cases, replay from pinned inputs, a second check and 22 supported propositions. Cases whose initial accounts were corrected are marked.

Figure 2. Findings in data provenance and transformation. **A**, Deposited molluscan occurrences present in (297,657) and absent from (71,103) the PBDB snapshot of 23 October 2026. **B**, Intervals with a changed modal base age in each of three snapshot pairs (at least 50 single-interval occurrences in the later snapshot). **C**, Full-precision change in mean absolute paleolatitude for the 2,267 comparable lineage-bins when excess Triton rows are removed. **D**, Stored minus interpolated NSB sample ages for the 56,387 comparable samples, with $\pm$0.01 Myr marked.

Figure 3. Findings in computation, model sensitivity and output interpretation. **A**, admtools 0.7.0 time error against height in the sparse synthetic model; vertical lines mark the two hiatuses. **B**, fossilbrush 1.0.6 flag decisions across the test grid, against numeric truth and text order, at a threshold of 5. **C**, Observed coverage against the 95% nominal line, for PyRate preservation-rate intervals (two arms) and for range-end bounds (three implementations). **D**, Median paleolatitude spread among three plate models by age, over the points reconstructable in all three at each age.

Tables

Table 1. The twelve cases. Families of check: DP, data provenance and transformation; CI, computation and implementation; MS, model sensitivity; OI, output interpretation and calibration; primary family first. Discovery: the route in the dated records. Review: what the replay and review trail changed in the initial account.

#	Resource	Family	Discovery	Check	Result	Review	Claims
1	Published PBDB deposit	DP	Source-membership audit; regional-support follow-up	Identifier comparison with a dated snapshot	71,103 of 368,760 absent; screen passes 14 then 12 of 21 stages	"withheld" became "absent from a dated snapshot"	C191, C192
2	PBDB interval ages	DP	Comparison of archived snapshots	Recount from archived query products	460, 317, 17 changed modal base ages	none	C076

#	Resource	Family	Discovery	Check	Result	Review	Claims
3	Triton 2023 update	DP	Duplicate-key profiling; exposure rebuild	Key grouping; rebuild with and without removal	1,388 of 2,267 means changed	counts re-computed	C043, C045
4	NSB stored ages	DP	Recomputation from tie points	Interpolation from tie points	4,111 of 56,387 differ by > 0.01 Myr	header row removed	C040
5	admtools 0.7.0	CI	Round-trip probe; declared hiatus benchmark	Exact linear-segment truth	3,972 of 8,282 heights mis-converted (sparse model)	"every hiatus" became adjoining segments	C096
6	fossilbrush 1.0.6	CI	Smoke-test failure; source inspection	Deterministic grid; numeric truth and text order	differences of 10, 20, 49.9, 100 and 499 unflagged at a threshold of 5	none	C102
7	PyRate preserva-tion rates	OI	Declared coverage benchmark	Simulation coverage, with positive controls	81/100 and 33/50 covered	"too narrow" became "shifted high"	C107, C194, C196, C105, C111
8	GRIWM, Rextinct 1.1.0	OI, CI	Smoke run (interval); repeated draws (random state)	Zero-error control; weighted-prefix formula; seed sentinels	interval reflects dating error; caller's random state overwritten	none	C110, C098
9	Three plate models	MS	Cross-route comparison; three-model follow-up	Reconstruction of shared points; geometry of two routes	median spread 2.9–13.6°	none	C125, C123
10	Published spread-sheets	DP	Format inspection; table-to-workbook comparison	Comparison with intact binary and chapter workbooks	2,967 of 6,562 histories differ numerically from the binary deposit; 220 shifted rows	cell count separated from history count	C035, C048
11	Printed probability	CI	Reading the printed test; exact calculation	Exact arithmetic, four routes	0.316 against 3.4×10^{-16}	none	C190
12	INDITEK-2.0 scores	CI	Declared forward replay; rescoring	Forward replay; separately written rescoring	vectors replay; saved scores differ from rescoring under deposited observations by up to 42.4	none	C195, C193

Table 2. Recommended actions, by audience.

Audience	Actions	Cases
Users	Record query dates and versions; check deposited records against the snapshot used; compare plate models and state the reference frame; replay a published number before building on it	1, 2, 9, 11
Database maintainers	Version interval ages and records, with change logs; document how stored values derive from their sources	1–4
Package maintainers	Test against known truth at boundary cases; compare numbers as numbers; leave the caller's random-number state intact; document what each interval measures	5–8
Journals and depositors	Require machine-readable deposits; archive scoring code and observations with saved scores	10–12
Programs using AI agents	Pair each finding with a second check; declare tests before running them and keep failed gates; publish receipts and reviewed correspondence	all